

Disclaimer

The U.S. Government does not endorse products or manufacturers. Trademarks or manufacturer's names appear in this presentation only because they are considered essential to the objective of the presentation. They are included for informational purposes only and are not intended to reflect a preference, approval, or endorsement of any one product or entity.

Except for any statutes and regulations cited, the contents of this presentation do not have the force and effect of law and are not meant to bind the States or the public in any way. This presentation is intended only to provide information regarding existing requirements under the law or agency policies.

AI Rapid Prototypes

Predictive Analytics for Incident Management

Session Name: AI for Proactive Road Safety
Date: August 19, 2026



U.S. Department of Transportation
ITS Joint Program Office

Source: iStock.com/Vertigo3d/cherzoff

AI Rapid Prototypes: Goal

CHALLENGE: Agencies and partners face challenges adopting and effectively using AI to support critical multimodal transportation needs

- AI changing rapidly
- Traditional research cycles often too slow
- Organizational challenges to deploying AI

GOAL: Demonstrate how AI can help meet critical transportation challenges through collaborative partnering, seed funding, and rapid prototyping

- Completed four prototypes, including **Incident Prediction prototype with WSDOT and MnDOT**

AI & Incident Management: Problem/Need

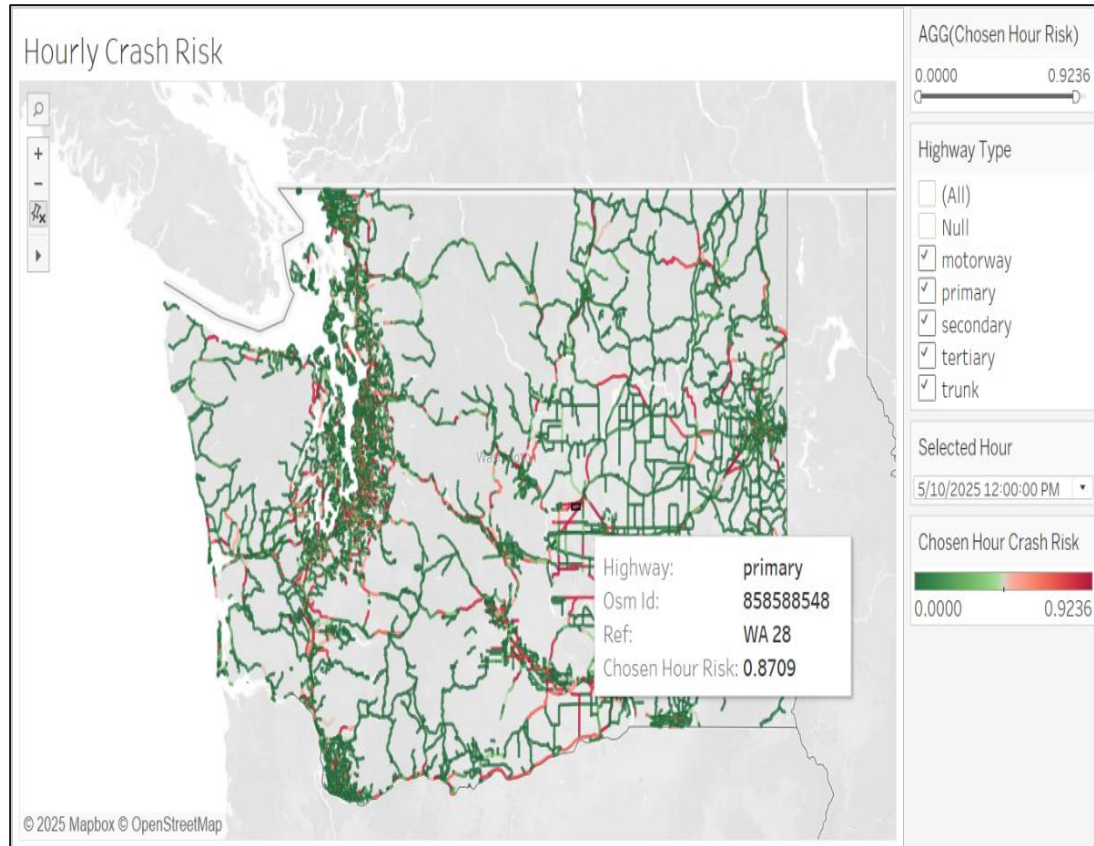
- Opportunities to explore how transportation agencies can transition from primarily **reactive** incident management to **proactive and predictive**
- Recognize different needs staff and leadership have for data and information
 - Transportation Management Centers (TMCs): **need second-to-minute** awareness
 - Safety teams: **need hourly-to-daily** forecasts
- Better predictive insights can leverage prevention opportunities and possibly **better safety outcomes**

AI & Incident Management: Approach

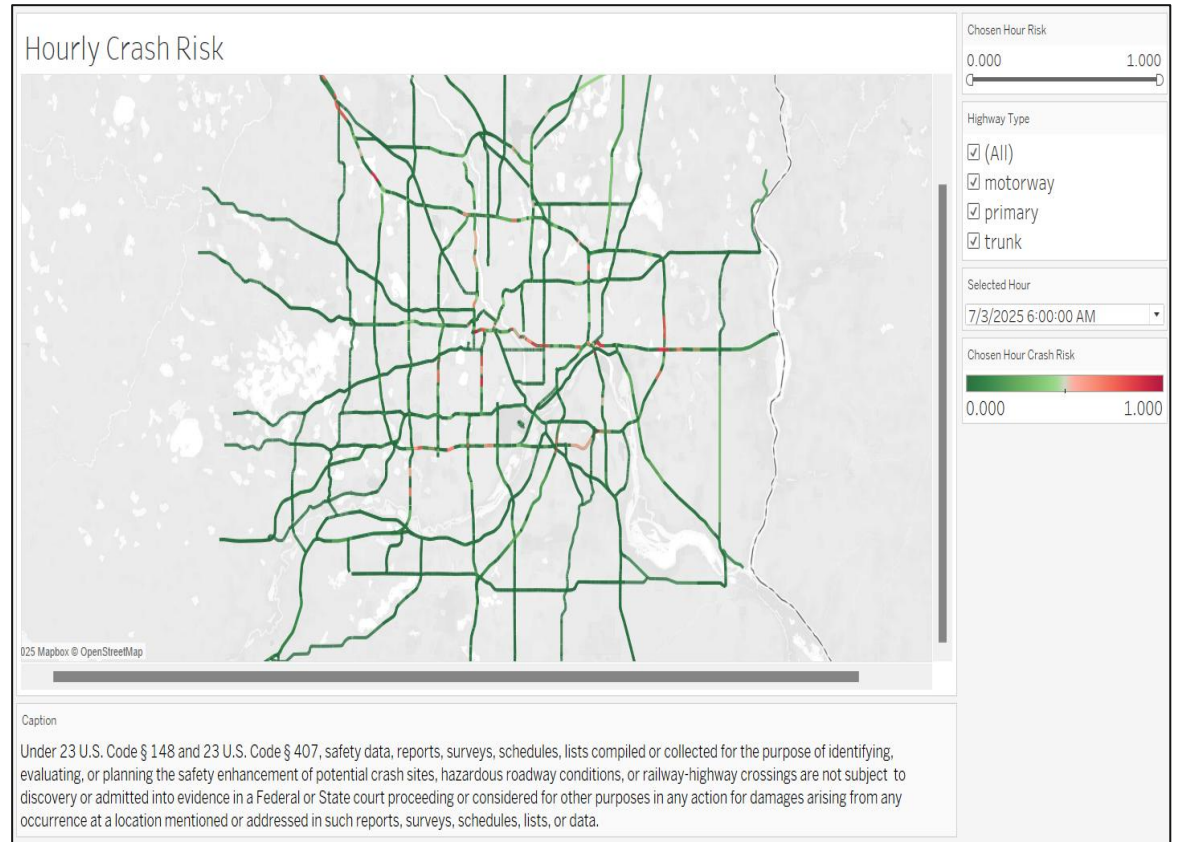
- **Fused data** (Waze, weather, historical crashes, and roadway data)
- Aggregated data into *hourly risk predictions*
- Created **interactive dashboard** with risk maps & filters
- Developed **natural-language assistant** for queries and visualization

AI & Incident Management: Crash Risks

Washington DOT: Hourly Crash Risks



Minnesota DOT: Hourly Crash Risks



Source: USDOT ITSJPO

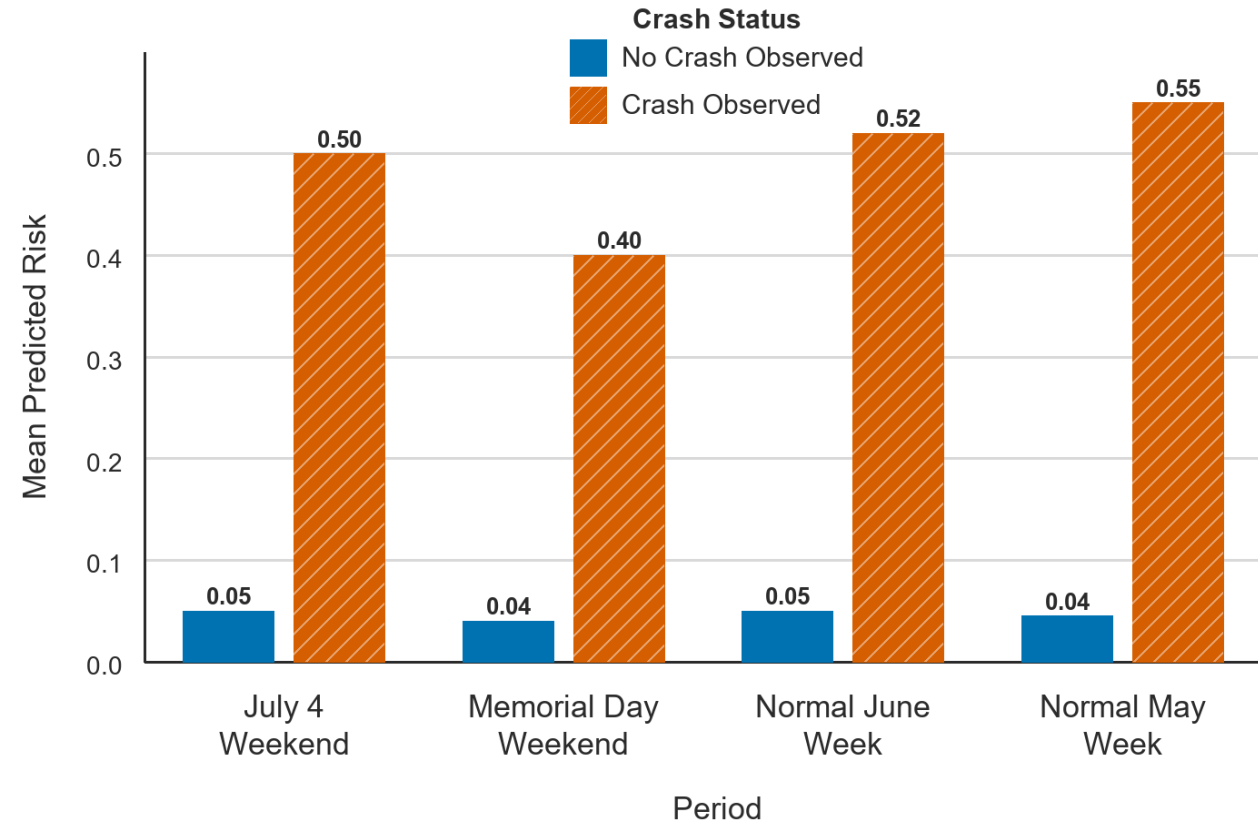
AI & Incident Management: MN Results

- **Accuracy:** Predicted high-risk corridors *12x to 22x better* than statistical benchmark methods during pilot weeks.
- **Peak Commuter Focus:** Delivered highest accuracy during active *weekdays and daytime hours (6 AM–6 PM)*.
- **High-Yield Route Coverage:** Performed best on *Interstate Highways, U.S. Routes, and State Routes*, where safety interventions have high impact.
- **Actionable Risk Rankings:** Ranks roads by safety risk rather than predicting rare, isolated events.

AI & Incident Management: MN Results

Impact of Holidays:
Models are more accurate during normal conditions, but can be finetuned to account for holidays, special events, and other factors.

Comparison of Mean Predicted Risk by Crash Status
(Minneapolis, MN; May – July 2025)



AI & Incident Management: WA Results

- **Validates TMC Expertise:** WSDOT Southwest Region operators confirmed that the *AI risk hot spots closely aligned* with their real-world experience.
- **Bridges Operator Gaps:** Serves as a *reliable guide* to known safety risk corridors for newer, less-experienced TMC staff.
- **Informs Capital Investments:** Can be leveraged for safety planners and engineers to *prioritize corridor-level infrastructure upgrades and countermeasures.*
- **Streamlines Public Inquiries:** Explored how GenAI can be used to respond to *public safety inquiries.*

AI & Incident Management: Key Takeaways

- **Importance of Human Decisions:** Key AI oversight role for operator staff who have quick-response decision needs. Important to *leverage staff expertise and experience*.
- **Critical to Integrate Predefined Decision Points:** Important to *link risk thresholds to TMC operational procedures* to realize full safety benefits.
- **Role-Tailored Interfaces:** Recognize needs of different agency staff and leadership (e.g., operators need high-priority alerts; planners can utilize in-depth corridor details).
- **Key Leadership Oversight:** Deployment success depends on early executive buy-in and regular engagement.

AI Prototypes: Partnering Opportunities

- **Established an AI Rapid Deployment Blueprint for USDOT:** Modular, reproducible approach can be expanded to other multimodal use cases. USDOT benefits from direct stakeholder feedback to inform larger Departmental AI initiatives.
- **Designed an Approach for Future Collaboration with State and Local Agencies:** Clear process established to identify, develop and deploy wide variety of priority multimodal AI use cases.
- **Open to Additional Ideas:** Low risk opportunity for partners to test AI. Partner stakeholder agencies receive a custom prototype for future use.

Thank you



U.S. Department of Transportation
ITS Joint Program Office

Source: iStock.com/Vertigo3d/cherzoff